

Using Deep Learning to Guide Phylogenetic Tree Search

Lena Collienne

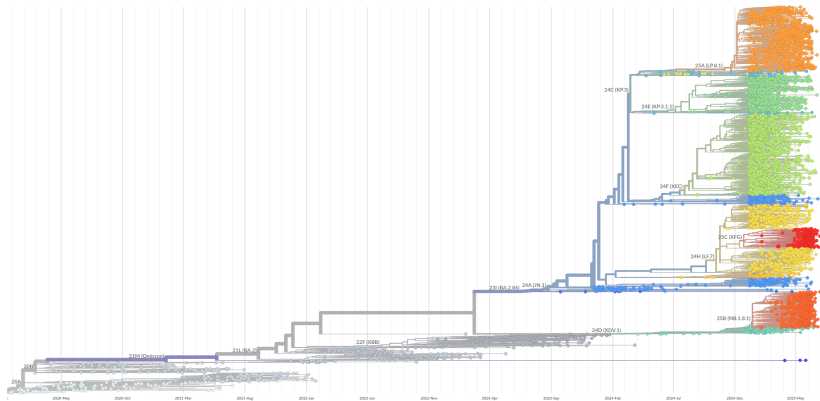
Matsen Group



Fred Hutch
Cancer Center

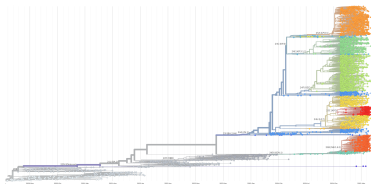
July 24th, 2025

Transmission Trees - SARS-CoV-2



<https://nextstrain.org/ncov/gisaid/global/6m>
(subsamped to 4435 samples)

Transmission Trees - SARS-CoV-2



Densely sampled data

- ▶ Large number of samples
- ▶ Very few mutations per branch

⇒ We can use Maximum Parsimony inference!

Maximum Parsimony

Sequence Alignment:

ACA

CAA

AAA

GTG

GGG

WANTED:

Simplest explanation
of evolution

Maximum Parsimony

Sequence Alignment:

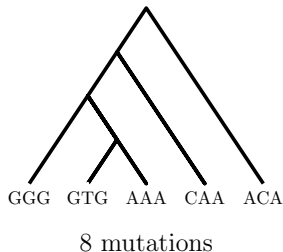
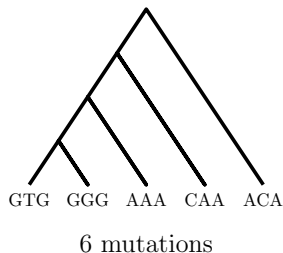
ACA

CAA

AAA

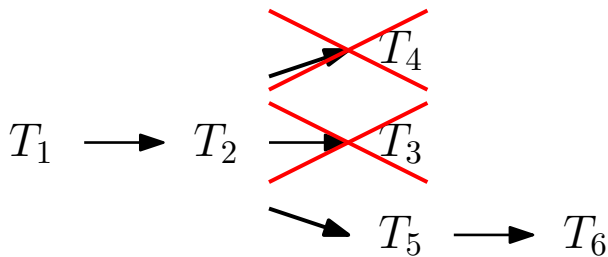
GTG

GGG



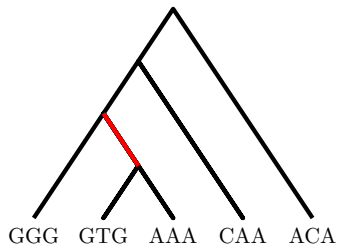
Maximum Parsimony

Tree Search

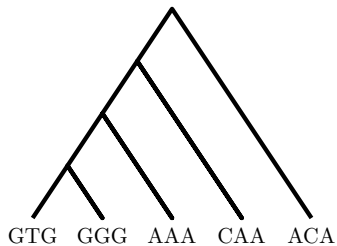


Maximum Parsimony

non-MP edges



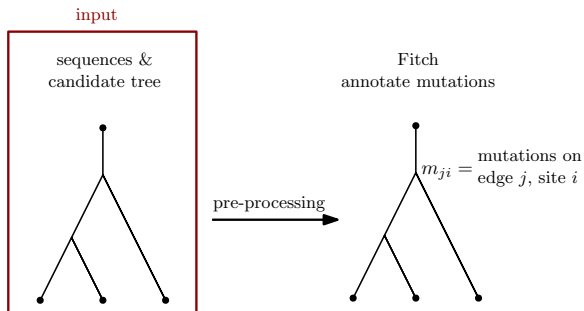
8 mutations



6 mutations

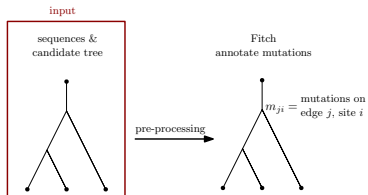
DPVT

Overview



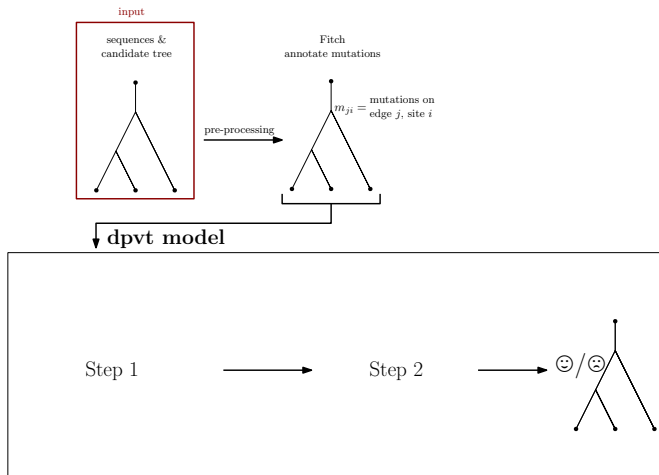
DPVT

Overview



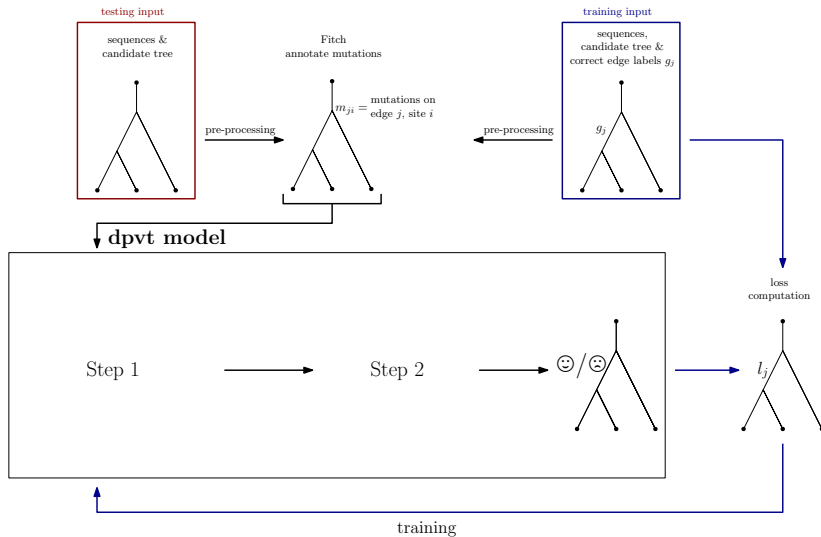
DPVT

Overview



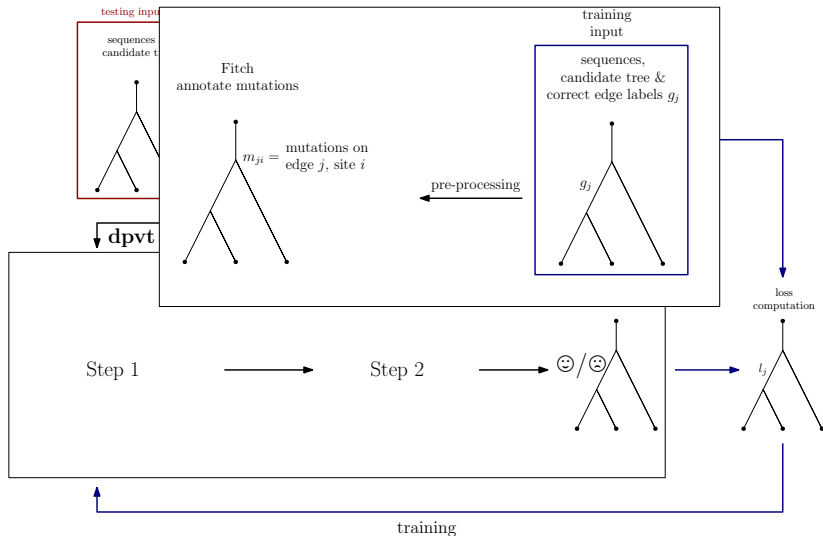
DPVT

Overview



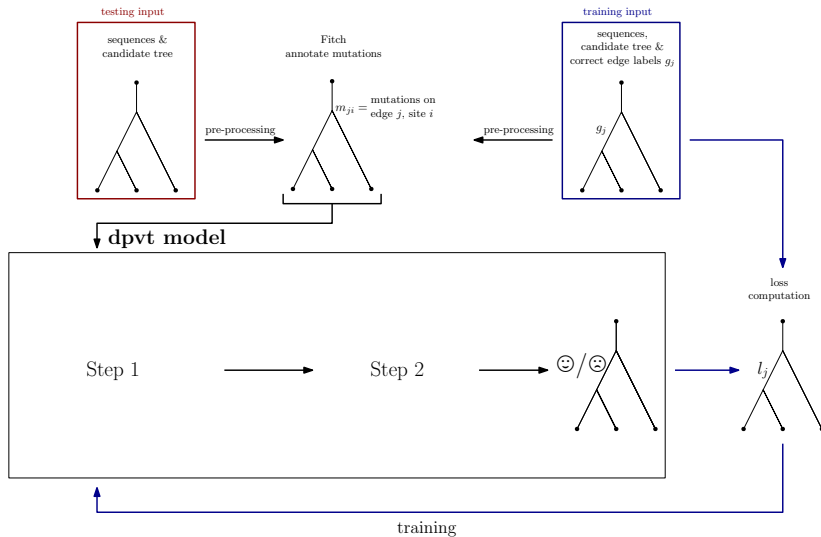
DPVT

Overview



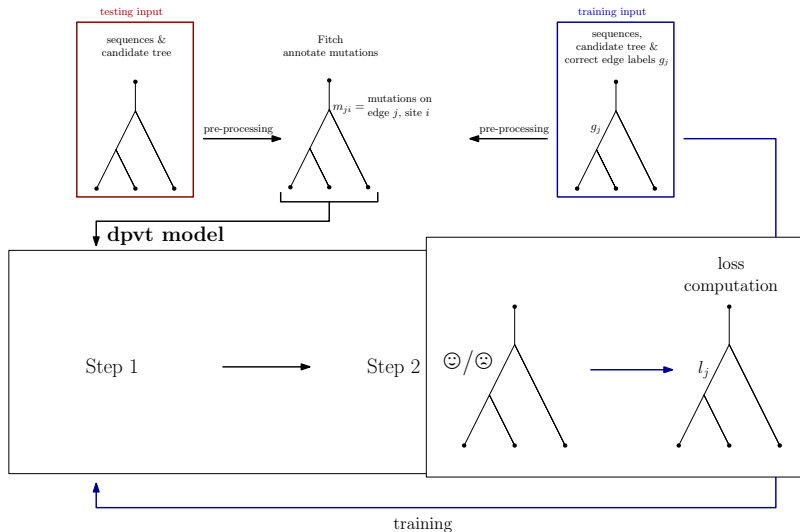
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Overview



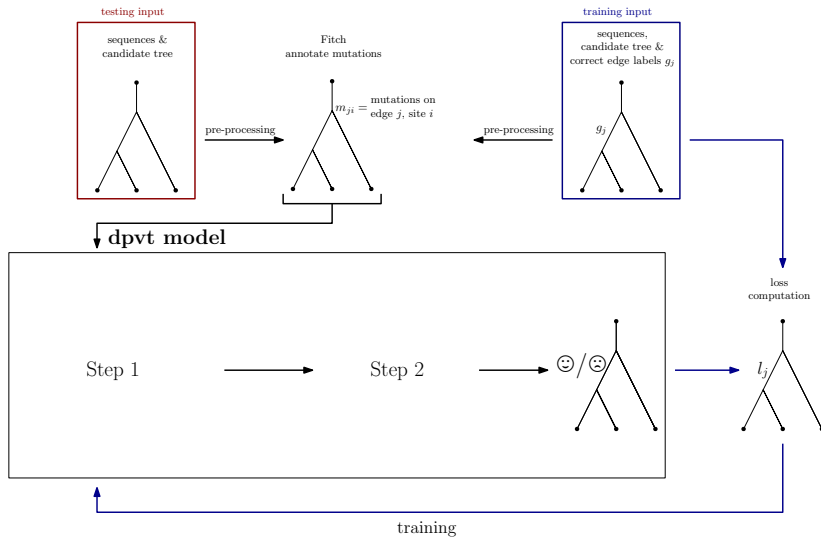
DPVT

Overview



DPVT

Overview



DPVT

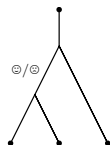
Overview

dpvt model

Step 1



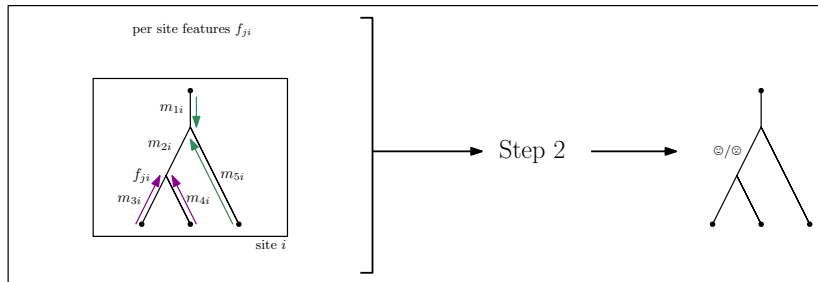
Step 2



DPVT

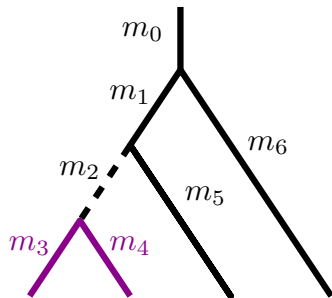
Overview

dpvt model



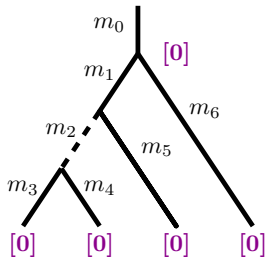
DPVT

Idea



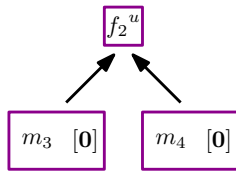
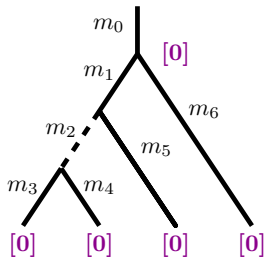
DPVT

Upward - per site



DPVT

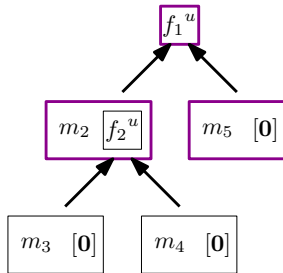
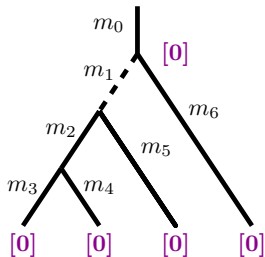
Upward - per site



$$f_2^u = r(m_3, [0], m_4, [0])$$

DPVT

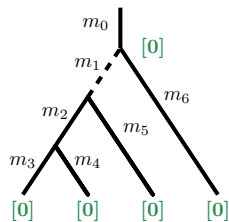
Upward - per site



$$f_1^u = r(m_2, [0], m_5, [0])$$

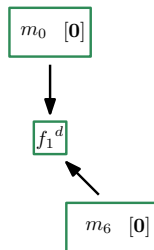
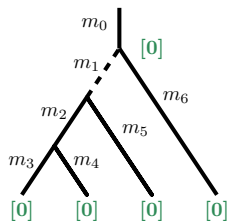
DPVT

Downward - per site



DPVT

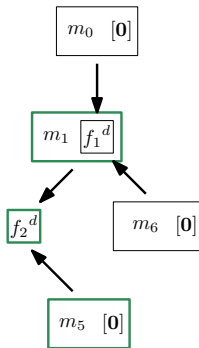
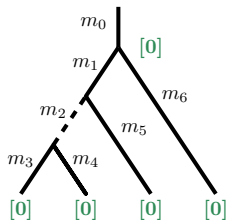
Downward - per site



$$f_1^d = r(m_0, [0], m_6, [0])$$

DPVT

Downward - per site

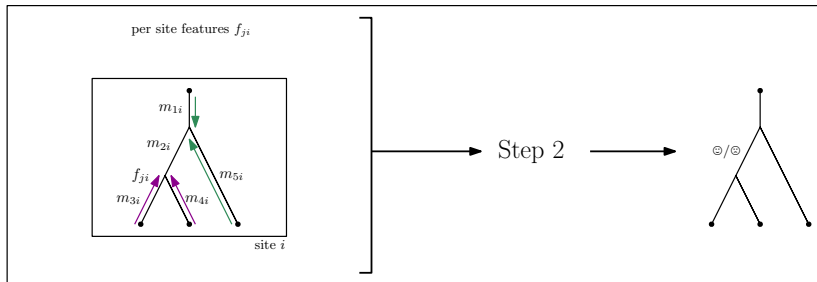


$$f_2^d = r(m_1, f_1^d, m_5, [0])$$

DPVT

Downward - per site

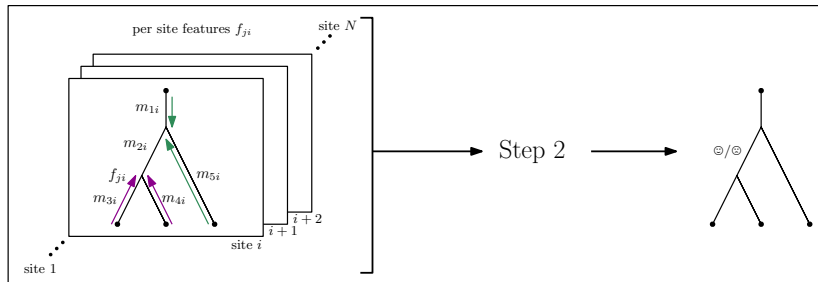
dpvt model



DPVT

Downward - per site

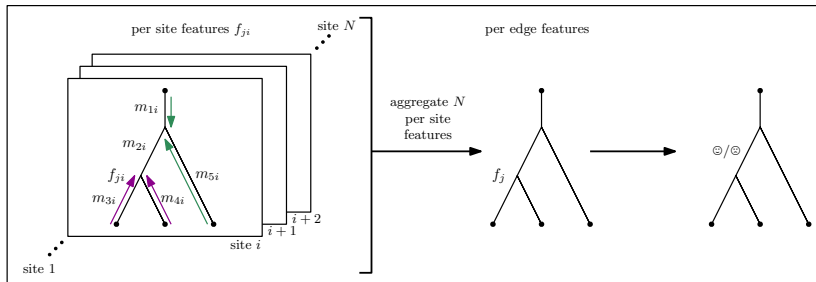
dpvt model



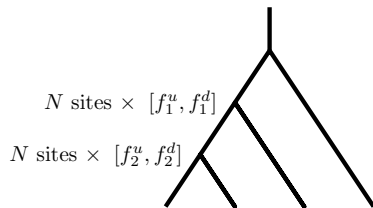
DPVT

Downward - per site

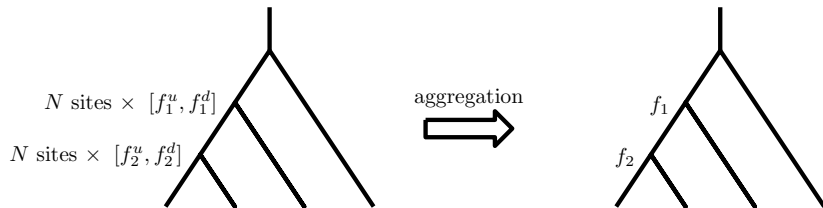
dpvt model



Aggregation over sites



Aggregation over sites

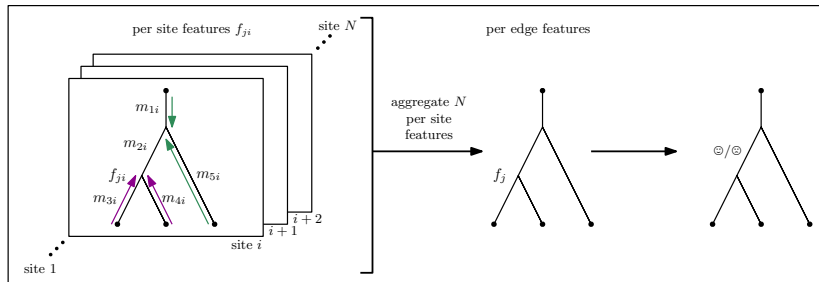


Different aggregation methods:

- ▶ Average
- ▶ Maximise
- ▶ Transformer Encoder

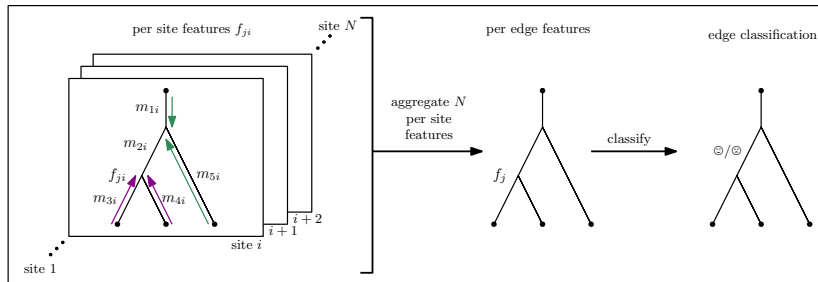
Aggregation over sites

dpvt model

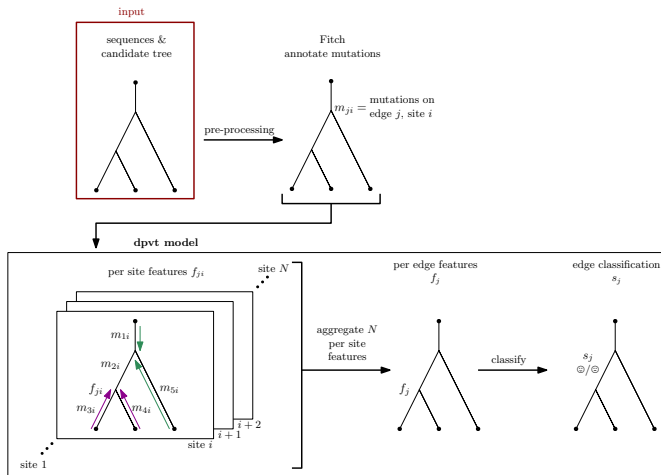


Aggregation over sites

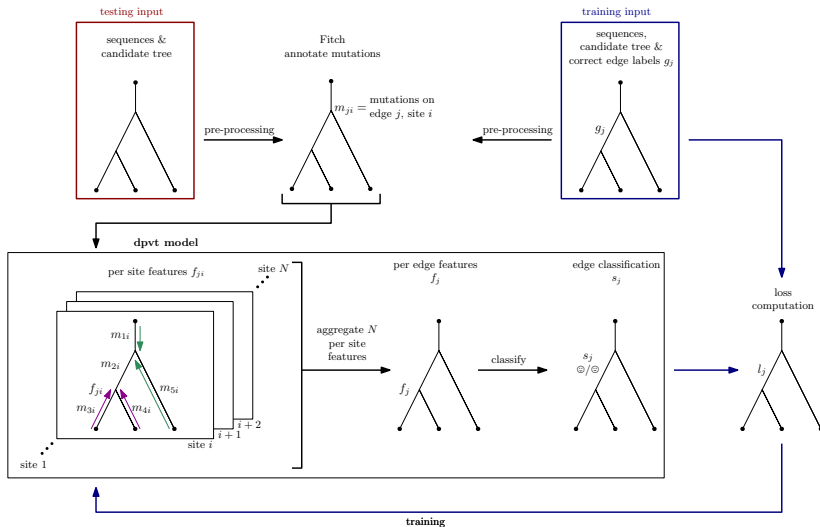
dpvt model



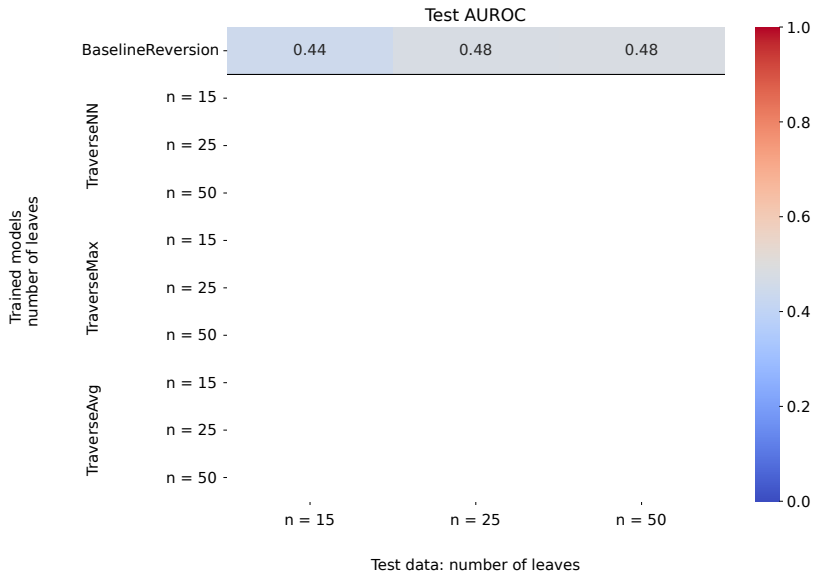
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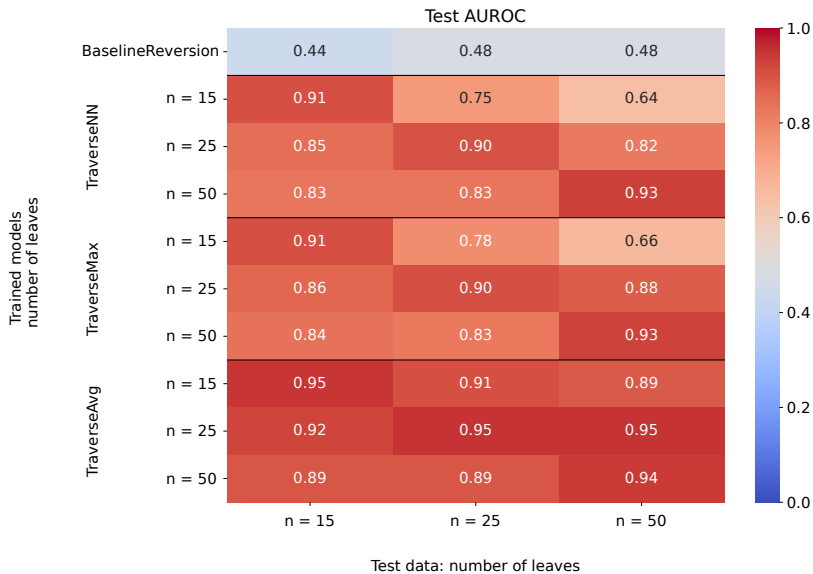
DPVT



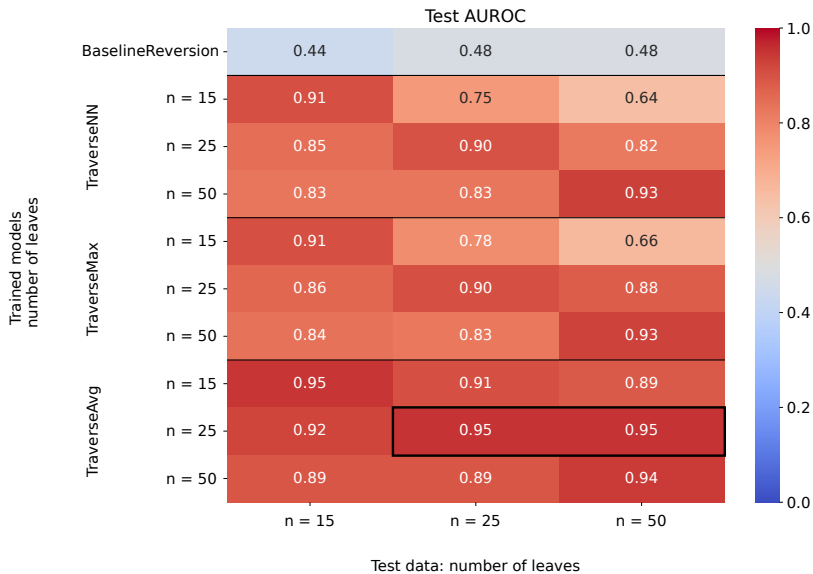
Results



Results



Results



Acknowledgements

Fred Hutch

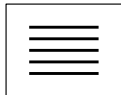
- ▶ Erick Matsen
- ▶ Harry Richman
- ▶ David Rich
- ▶ Chris Jennings-Shaffer
- ▶ Mary Barker



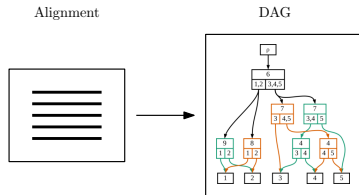
Appendix

Training Data

Alignment

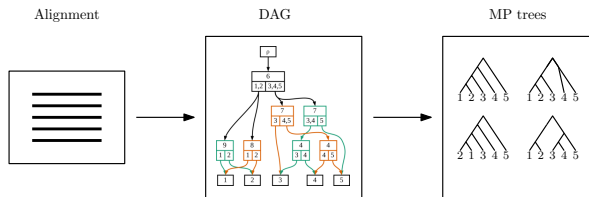


Training Data



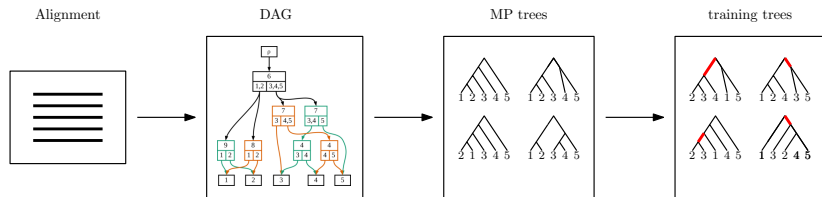
DAG figure from: Dumm et al 2023

Training Data



DAG figure from: Dumm et al 2023

Training Data



DAG figure from: Dumm et al 2023